



# The 66th ASH Annual Meeting Abstracts

## POSTER ABSTRACTS

### 722.ALLOGENEIC TRANSPLANTATION: ACUTE AND CHRONIC GVHD AND IMMUNE RECONSTITUTION

#### Pooled Fecal Allogenic Microbiotherapy for Refractory Gastrointestinal Acute Graft-Versus-Host Disease : Results from Early Access Program in Europe

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## Introduction

Acute graft-versus-host disease (aGvHD) is a major source of mortality following allogeneic hematopoietic cell transplantation (allo-HCT). Fecal microbiotherapy has shown promising results in several pilot studies in patients with refractory gastrointestinal (GI)-aGvHD. Here we report long-term clinical outcomes of 154 patients diagnosed with refractory GI-GvHD treated with the pooled allogeneic microbiotherapy MaaT013 within an Early Access Program (EAP) in Europe.

## Patients and methods

One hundred and fifty four patients (incl. two pediatric patients aged 12 and 15 years) with steroid-refractory (SR) or -dependent (SD) GI-aGvHD (classical n=93, late onset n=16, chronic or overlap syndrome n=25, hyperacute n=20) were treated with MaaT013 as part of the EAP in Europe (France, Germany, Spain, Italy, Austria). These patients had previously failed 1 to 6 systemic aGvHD treatment lines (median 3; 134/152 received ruxolitinib). Most patients had grade III to IV aGvHD (13% grade II, 47% grade III, 40% grade IV).

For each patient, a total of 3 MaaT013 administrations were planned every 7 +/- 2 days (median dose administered 3, range 1-3). Each dose comprised 30 g of feces in 150 mL of solution from 4 to 8 healthy donors administered by enema (except for 1 patient by nasogastric tube).

Treatment response was calculated among treated patients based on aGvHD grading at day 28 (D28) compared grading at the time of the EAP request.

## Results

At D28, the GI-overall response rate (ORR) was 51%: 46 complete response (CR 30%), 23 very good partial response (VGPR 15%), 9 partial response (PR 6%). GI-ORR was higher in grade II & III GvHD patients compared to grade IV (65% in grade II, 64% in grade III, 30% in grade IV) and higher in SD versus SR (81% versus 45%). ORR considering all organs (n=154) was 49%, including 41 CR, 19 VGPR and 15 PR.

At D56, GI-ORR was 44% with 50 CR, 13 VGPR and 3 PR, ORR was 42% with 47 CR, 12 VGPR and 5 PR (n=151).

Administration of MaaT013 in the two pediatric patients was well tolerated. One patient was in VGPR at D28 (baseline grade IV, stage 0 skin and liver, stage 4 GI-aGVHD), the other was in complete response of GI and skin symptoms up to 12 months (baseline grade IV, stage 3 skin, stage 2 liver, stage 4 GI) but liver symptoms were not resolved at D28 (stage 2 liver, stable disease). Liver stage improved from month 6 (M6) without additional therapy.

Overall survival (OS) was 53% at M6, 47% at M12, and 42% at M24. The median follow-up among surviving patients was 418 days (range, 27-1644). OS was significantly higher in patients achieving at least GI-PR at D28 (Responder, R; n=78) compared to patients in treatment failure (Non-responder, NR; n=76): 73% versus 34% at M6, 68% versus 24% at M12 and 58% versus 24% at M24 (Log-rank test p<0.0001). Median survival duration in R was 290 days versus 71 days in NR.

Interestingly, in the subgroup of 58 patients previously treated with ruxolitinib as 2<sup>nd</sup> line and MaaT013 as 3<sup>rd</sup> line GI-ORR was improved being 59% at D28 (48% CR) and 54% at D56 (53% CR). ORR was 55% (43% CR) at D28 and 56% (53% CR) at D56. OS was 54% at M6, 49% at M12, and 40% at M24. OS was significantly higher in R patients, when compared to NR patients (79% versus 17% at M6, 75% versus 11% at M12, 61% versus 11% at M24) for R and NR, respectively, Log-rank test p<0.0001. Median survival duration in R was 444 days versus 42 days in NR.

MaaT013 displayed a good overall safety profile in the EAP population: 37 serious pharmacovigilance cases were reported in 34 patients, including 24 cases reported in 23 patients considered possibly related to MaaT013 either by the physician or the company: infections/ bacteremia in 13 patients, sepsis in 6, rectal bleeding/ anorectal disorder in 3, *C. difficile* colitis in 1.

83 deaths have been reported. The causes of death were GvHD in 34 patients, severe infection in 30 (incl 5 COVID-19), relapse of underlying malignancy in 11, hemorrhage in 2, neurological complications in 2, cardiac arrest in 2, acute respiratory distress in 1 and natural death in 1.

## Conclusion

Overall, EAP clinical data showed that MaaT013 was a safe and effective treatment of refractory GI-aGvHD especially in patients having previously received ruxolitinib. Interestingly, the response to MaaT013 correlates with increased OS, suggesting a strong favorable benefit-risk profile for MaaT013. A Phase 3 trial is currently ongoing to confirm these results in ruxolitinib-refractory patients (NCT04769895).

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